Lesson 7.10 — Solving Trigonometric Equations

Big idea: Solving a trig equation = isolating sin/cos/tan, then finding ALL angles in the requested interval that satisfy it. Don't forget all solutions in $[0, 2\pi)$ or all infinite solutions "$\theta = \dots + 2\pi n$."

Quick review

Step 1: isolate the trig function.
Step 2: use the inverse to get a reference angle (calculator).
Step 3: find all angles in the requested interval using quadrant logic.
Step 4: for periodic infinite answers, add $2\pi n$ (or πn for tangent).
Watch: squared trig terms or substitutions like $u = \sin\theta$; check for extraneous solutions.

Worked example

Worked example. Solve $2\sin\theta - 1 = 0$ for $0 \le \theta < 2\pi$.

$\sin\theta = 1/2$. Reference angle $\pi/6$. Q1: $\theta = \pi/6$. Q2: $\theta = \pi - \pi/6 = 5\pi/6$. Solutions: $\pi/6$, $5\pi/6$.

Warm-ups

1. Solve $\sin\theta = 0$ for $0 \le \theta < 2\pi$.
2. Solve $\cos\theta = 1$ for $0 \le \theta < 2\pi$.
3. Solve $\tan\theta = 0$ for $0 \le \theta < 2\pi$.

Practice

1. Solve $2\cos\theta + 1 = 0$ for $0 \le \theta < 2\pi$.
2. Solve $\sin\theta = \sqrt{2}/2$ for $0 \le \theta < 2\pi$.
3. Solve $\tan\theta = -1$ for $0 \le \theta < 2\pi$.

Challenge

1. Solve $2\sin^2\theta - \sin\theta - 1 = 0$ for $0 \le \theta < 2\pi$. (Substitute $u = \sin\theta$.)
2. Solve $\sin(2\theta) = 1/2$ for $0 \le \theta < 2\pi$.

Answer key — Lesson 7.10

Check your work. If a problem tripped you up, rewatch the step in the video where that concept appears.

Warm-ups

1. $\theta = 0, \pi$.
2. $\theta = 0$.
3. $\theta = 0, \pi$.

Practice

1. $\cos\theta = -1/2$. Reference $\pi/3$. Q2: $2\pi/3$, Q3: $4\pi/3$.
2. Reference $\pi/4$. Q1: $\pi/4$, Q2: $3\pi/4$.
3. Reference $\pi/4$. Q2: $3\pi/4$, Q4: $7\pi/4$.

Challenge

1. $(2u + 1)(u - 1) = 0 \Rightarrow u = -1/2$ or $u = 1$. $\sin\theta = 1 \Rightarrow \theta = \pi/2$. $\sin\theta = -1/2$: reference $\pi/6$, Q3: $7\pi/6$, Q4: $11\pi/6$. So $\theta = \pi/2, 7\pi/6, 11\pi/6$.
2. Let $\phi = 2\theta$, so $0 \le \phi < 4\pi$. $\sin\phi = 1/2 \Rightarrow \phi = \pi/6, 5\pi/6, \pi/6 + 2\pi = 13\pi/6, 5\pi/6 + 2\pi = 17\pi/6$. So $\theta = \pi/12, 5\pi/12, 13\pi/12, 17\pi/12$.